Fiscal Year:	FY 2026	**Task Last Updated:**	FY 07/15/2025
PI Name:	Zupanska, Agata Ph.D.		
Project Title:	From Antarctica to Space: Molecular Response and Physiological Adaptation of Moss to Simulated Deep Space Cosmic Ionizing Radiation and Spaceflight Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	94043-5138	**Congressional District:**	18
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/21/2025 per the PI (Ed., 6/9/25).		
Key Personnel Changes/Previous PI:	As budgeted in the original proposal we hired an OPS Technician, Emily Lockwood. Emily is affiliated with University of Florida and paid from the subaward. Emily works on the project accordingly to proposed effort, 6 months for each year of grant project.		
COI Name (Institution):	McDaniel, Stuart Ph.D. (University of Florida, Gainesville)		
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Task Description:	Rationale: Deep space environments of cosmic ionizing radiation (cIR) from astrophysical events and microgravity (micro-g) are outside of plant evolutionary experience and can impede the vital role of plants in space exploration. Understanding the combined effects of cIR and micro-g on plants at the molecular level is pivotal to engineering deep space-tolerant plants for human space exploration. Most previous research studies each condition alone and uses seeds irradiated with high doses of acute cIR. Hypothesis and Objectives: We propose a Research Investigation (Project Type) into plant growth, development, and physiology (Focus Area) using an International Space Station (ISS)-Flight+Ground Study Component (Platform) to test the hypothesis that combined deep space cIR and micro-g affect plants differently and more profoundly than each environment alone, resulting in distinct molecular responses and physiological adaptation. We aim to detect how the effects of combined simulated chronic deep space cIR and sequential spaceflight micro-g differ from those of cIR or micro-g alone on live plant growth, development, and physiology, as reflected by changes in gene expression, to identify molecular response and physiological adaptation to these environments. Research Approaches and Specific Aims: We will use Antarctic moss Ceratodon purpureus (C. purpureus) which allows us to use irradiated live plants rather than spores in spaceflight. Ceratodon purpureus is tolerant of extreme conditions and is ideal for human spaceflight missions due to its small size, lightweight spores, low-maintenance, and roles in oxygenation, regolith enrichment, and medicine production. We will use a twofold approach: 1) in the ground study protonema will be exposed to simulated Galactic Cosmic Rays (GCRSim, 10 and 100 cGy) and Solar Particle Events (SPESim, 100-1000 cGy) delivered at fractionated dose; 2) in the ISS-Flight study protonema exposed to GCRSim or SPESim on the ground and housed in the NASA Biological Research in Canisters (BRIC)-Light Emitting Diode (LED) will be launched to the ISS and grown in low-Earth orbit (LEO). We will use morphological metrics, RNAseq, and bioinformatics to achieve: AIM1- Discover the dose- and time-dependent effects of simulated deep space cIR on growth, development, and gene expression in C. purpureus. AIM2- Identify the different effects on growth, development, and gene expression in C. purpureus of simulated deep space cIR – singular, or combined with spaceflight micro-g. AIM3- Identify the different effects on growth, development, and gene expression in C. purpureus of spaceflight micro-g – singular, or combined with simulated deep space cIR. Outcome and Significance: This study will: 1) analyze moss growth and development in cIR and micro-g, which will inform plant performance capabilities for human spaceflight exploration; 2) discover moss response and physiological adaptation to cIR and micro-g at the gene expression level, and by extension in other biological systems; 3) discover candidate genes for genetic manipulation to engineer deep-space tolerant plants which will advance the design of Bioregenerative Life Support systems and plant habitats for planetary missions; and 4) extend our results to other multicellular organisms, which will inform health risk management. Relevance to the Plant Biology Element of the NASA Space Biology program: This project addresses: a) the main goal, to “understand plant growth in spaceflight environments and physiological responses to those environments;” b) guiding questions #1 - How does gravity affect plant growth, development and metabolism; #4 - What are the effects of chronic exposure to cosmic radiation on plants; and #5 How do plants sense and react to gravity and what are the molecular mechanisms involved; and c) Plans and Research Solicitations objectives #1 Determine how plants adapt to spaceflight from the molecular to whole plant level using existing hardware (BRIC) on ISS; #3 - Answer basic research questions about plant biology that will enable the future development of a bioregenerative life support system for use by humans in space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In our study we deploy novel approaches in plant space biology research enabling to maximize scientific return. First, unlike other studies relying on model plants, we bring a new organism, moss, which is remarkably adaptive to extreme environments here on Earth and thus is likely to exhibit superior adaptability to deep space environments. Second, in contrast to most other studies which address each deep space environment separately, we will discover the combined effects of both cosmic ionizing radiation and spaceflight microgravity. Third, in contrast with previous research done on irradiated seeds and spores which leads to mutagenesis rather than observation of the effects of deep space cosmic ionizing radiation on plant physiology, our pioneering experimental design sequentially exposes live, actively growing plants to deep space environments. Fourth, we deploy the most accurate simulation of cosmic ionizing radiation of astrophysical origin, exposure to ion beams of compositions and energies best approximating Galactic Cosmic Rays (GCR) and Solar Particle Events (SPE) radiation. Most studies used exposure to electromagnetic wave radiation from UVB, UVC, X-rays, and Gamma-rays delivered at a high-dose in a single pulse (acutely), which poorly simulates deep space cosmic ionizing radiation, which is particulate, chronic, and low-dose at extended fractionations. Upon completion we will deliver moss transcriptomic profiles, growth and development analysis and candidate genes for genetic manipulation to engineer plants more tolerant of cosmic ionizing radiation and spaceflight microgravity. Firstly, this will have a direct impact on advancing plant preparedness for space exploration missions and will help constructing plant varieties that can be incorporated into life support systems and in-space human habitation, enabling long-duration crewed missions beyond Low Earth Orbit. Secondly, since our study will identify the critical plant mechanisms of prevention or repair of the damage caused by ionizing radiation it will potentially result in developing moss-based radiation countermeasures agents of three classes: protectors, mitigators and therapeutics. Such plant compounds could be then used by astronauts during long term interplanetary travel as well as administered to radiation oncology patients or nuclear plant workers accidentally exposed to high radiation doses here on Erath.

Task Progress:

Key achievement: We successfully launched the "ANT1 Radiation Tolerance Experiment with Moss in Orbit on the Space Station" (ARTEMOSS) experiment to the International Space Station (ISS) – the first study to investigate Antarctic moss under the combined effects of cosmic ionizing radiation and microgravity. This marks the first time a plant native to Antarctica has been flown to the ISS, and the first instance where living plants, rather than seeds, were exposed on Earth to the most accurate simulation of cosmic ionizing radiation before experiencing microgravity in space.
1) SPECIFIC OBJECTIVES

Our objective was to sequentially expose Antarctic moss to simulated cosmic ionizing radiation (ion beam exposures at NSRL/BNL) and to spaceflight microgravity (growth on the ISS) to investigate the effects of those two space environments, alone and in combination, on moss growth and physiology as reflected by changes in gene expression.

We successfully performed the following tasks to achieve our objective:

1. Expose moss to ion beams at NASA Space Radiation Laboratory/Brookhaven National Laboratory (NSRL/BNL) as close to the launch date as possible: thanks to the generosity and dedication of the NSRL/BNL team, we managed to conduct the exposures five (5) days before the launch date.

2. Maintain moss under cold conditions (0-4°C) before, during, and post-exposures, through travel to the launch site, launch, orbit, ISS berthing): frozen ice packs during exposures, pre-chilled refrigerator, car refrigerator, pre-chilled growth chambers, cold stowage assured by SpaceX team all assured that moss temperature remained below 4°C with a brief period when it reached 5.9°C.

3. Conduct the ARTEMOSS experiment to the ISS: the ARTEMOSS was launched, flight operations were completed, moss tissue was effectively fixed in orbit through deep-freeze, and the samples were safely returned.

4. Conduct the SINGULAR IRRADIATION experiment alongside the ARTEMOSS experiment: the SINGULAR IRRADIATION experiment was completed, and sixty-six (66) samples were generated.

5. Extract enough RNA from moss samples, preserving as much tissue as possible: only four (4) moss colonies out of thirteen (13) were used for RNA extractions.

6. Obtain enough RNA of excellent quality: all samples met excellent quantity and quality success criteria.

7. Perform morphological analysis: nearly 200 images were collected and are currently under analysis.

8. Perform RNAseq analysis: RNAseq libraries were generated, and the next-gen sequencing was successfully finalized, and the data is currently under analysis.

2) DETAILED ACTIVITIES

• Exposure to ion beams at NSRL/BNL

Antarctic moss grown on Petri plates was exposed to ion beams at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL) five (5) days before the Dragon launch. The exposures simulated radiation encountered in deep space, including Solar Particle Event radiation (SPESim) at a dose of 40cGy and 200cGy, and Galactic Cosmic Ray radiation (GCRSim) at a dose of 20cGy and 80cGy. Moss was maintained under cold conditions before, during, and after exposures, through ground transport from NY to FL, payload integration, turnover, launch, orbit, and ISS berthing.

• ARTEMOSS payload

The 3-plate stacks used in ARTEMOSS were designed so that each of the three (3) biological replicates per radiation treatment occupied a different position (top, middle, or bottom), to mitigate location bias. At the NASA Space Station Processing Facility at Kennedy Space Center (KSC), five (5) such stacks for both ARTEMOSS FLIGHT and GROUND CONTROL were integrated into the ARTEMOSS science bag payload. A HOBO data logger was activated and placed next to the plate stacks. The integrated science bag was inserted into a flight-approved plastic bag that was then heat-sealed using a heat-sealer. The fully integrated ARTEMOSS payload was turned over to the SpaceX cold stowage team.

• ARTEMOSS spaceflight experiment

The ARTEMOSS spaceflight experiment was successfully launched aboard the NASA SpaceX CRS-31 mission on 11/04/2024. After the SpaceX Dragon docked with the ISS, the ARTEMOSS payload was removed from cold stowage and placed at ambient ISS temperature, initiating a seven (7)-day growth period. The ARTEMOSS GROUND CONTROL was initiated with a delay by placing the corresponding Petri plates in the ISSES chamber (KSC). After seven (7) days, the five (5) 3-plate stacks were removed from the ARTEMOSS science bag and transferred to a -150 °C freezer to terminate growth, then moved to a -80 °C freezer the following day. On 12/17/2024, the SpaceX Dragon splashed down off the Florida coast, and the Principal Investigator (PI) received the samples two (2) days later.

• ARTEMOSS sample processing

All ARTEMOSS samples were processed with strict care to keep moss tissue frozen throughout. Moss on each frozen Petri plate was inspected and photographed for the morphological analysis. One-third (1/3) of the tissue was harvested by scraping frozen moss from the media surface for RNA extraction. Extracted RNA was assessed for quality and quantity using a Bioanalyzer (Agilent), then used for RNA-seq library construction (Illumina RNA-seq libraries (Poly A)) with 250ng of input total RNA, followed by next-gen sequencing (Illumina NovaSeq X Series: 10B: 2x150 (1 lane)) with 30-50mln reads coverage.

• Data analysis

The RNAseq library was completed in January 2025, and the next-gen sequencing was finalized in February 2025. Imagery data and transcriptomic data is currently under analysis.

3) KEY OUTCOMES

Every sample from the ARTEMOSS experiment scored excellent in Science Performance in each of the Success Criteria:

1. Plant/moss viability – Moss remained viable throughout the entire ARTEMOSS experiment in spaceflight (FLT) and

	on the ground (GC), as reflected in the healthy green color of every colony on every Petri plate throughout. Importantly, the moss tissue was visibly green, and a substantial elongation of upward filaments was visible on the frozen Petri plates at the end of the ARTEMOSS experiment. 2. Plant/moss morphology – All moss colonies remained green with no signs of morphological aberrations at every ARTEMOSS step. No colony on any plate showed signs of browning, yellowing, or color bleaching. 3. Bacterial or fungal contamination – There was no bacterial or fungal contamination on any spot-inoculum, any Petri plate, at any point throughout the ARTEMOSS experiment. 4. Quantity of Total RNA extracted – More than 400ng of total RNA was extracted from four (4) moss colonies from each Petri plate. 5. Quality of Total RNA extracted – RNA Integrity Number (RIN) for all samples exceeded 7.0. In each sample, the 25S rRNA peak dominated and was clearly separated from the 18S rRNA peak, and the 25-nt-long RNA corresponding to small RNA contributed a minor fraction of the total RNA. There was no sign of RNA degradation in any sample. 4) MITIGATED OBSTACLES AND RISKS Obstacle: Ineffective chemical fixation of Antarctic moss tissue. We replaced chemical fixation with RNAlater by a deep-freezing method (at -150°C). Because of a different fixation method, we did not use Biological Research in Canisters / Petri Dish Fixation Unit (BRIC/PDFU) hardware, but organized our payload in 3-plate stacks, wrapped in aluminum foil, and housed in a Ziploc bag. Risk 1: Maintaining Petri plates at 0-4°C from the target room (NSRL/BNL) until the Petri plates are on the ISS. We have excelled in maintaining moss on Petri plates in cold conditions from the time of exposure to ion beams to the time when our payload was transferred to the ISS. - Prior to exposures, Petri plates were stored in the fridge at 1°C and were removed from the fridge moments before exposures; - During exposures, Petri plates resided on ice packs; - During transport from BNL/NY to KSC/FL, Petri plates were kept in a car fridge maintaining the temperature between 0-2.9°C; - After payload integration at KSC, the Petri plates were stored in a growth chamber maintaining 4°C; - At the turnover, the SpaceX stowage team assured the payload stowage in cold conditions. Although the temperature during transit to the ISS briefly reached 5.9°C, this temperature can still be considered low and is sufficient to inhibit moss metabolic activity. Risk 2: Synchronizing exposure to ion beams at BNL with the SpaceX-31 launch date. Although the launch date was repeatedly postponed, from June to August, then September, October, and finally to November 4th, making it impossible to schedule ion beam exposures in advance, we were able to conduct the exposures just before launch as desired. This was made possible thanks to the gracious support and exceptional dedication of the NSRL/BNL team, who accommodated us with overnight exposures outside their planned schedule.
Bibliography Type:	Description: (Last Updated: 07/17/2025)
NASA Technical Documents	Zupanska A. "ARTEMOSS. From Antarctica to Space: Molecular Response and Physiological Adaptation of Moss to Simulated Deep Space Cosmic Ionizing Radiation and Spaceflight Microgravity." NASA Space Station Research Explorer. 2024. https://www.nasa.gov/mission/station/research-explorer/investigation/?#id=9228 , Oct-2024
Significant Media Coverage	Gaskill ML. "NASA's SpaceX 31st resupply mission to launch experiments to Station. Description of Dr. Zupanska's spaceflight experiment, ARTEMOSS." Web feature article on nasa.gov. October 17, 2024. https://www.nasa.gov/missions/station/iss-research/nasas-spacex-31st-resupply-mission-to-launch-experiments-to-station/ , Oct-2024
Significant Media Coverage	O'Neal P. (Zupanska A interview). "SpaceX CRS-31 prelaunch science webinar." NASA and ISS National Lab hosted webinar, October 23, 2024. https://www.youtube.com/watch?v=LrpX8eX3nNg , Oct-2024